

Supplemental Information

Quantifying and mitigating motor phenotypes induced by antisense oligonucleotides in the central nervous system

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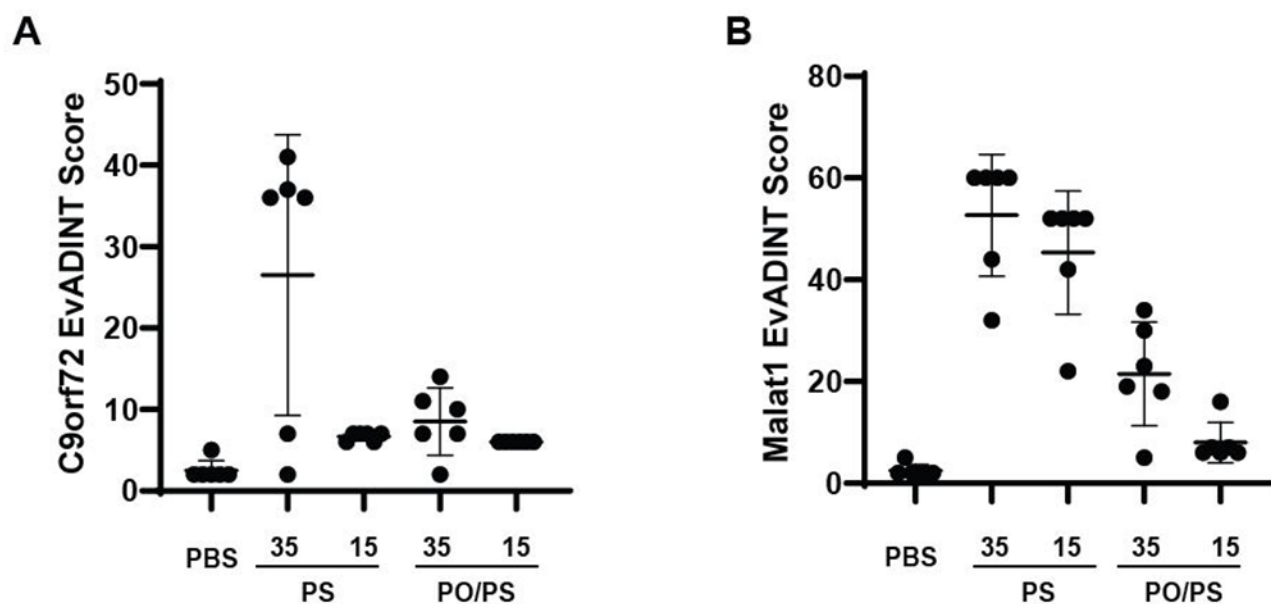


Figure S1. The acute neurotoxicity of ASOs is dose-dependent. Mice were injected ICV with 35 or 15 nmol of ASOs targeting (A) C9orf72 or (B) Malat1 in 10 μ L PBS (or with 10 μ L PBS as control) and behavior was scored by a blinded investigator over the following 24h using the EvADINT rubric. PS: fully phosphorothioate backbone, PO/PS: mixed backbone. Each data point represents the EvADINT score from one mouse; $n = 6$; error bars represent SEM.

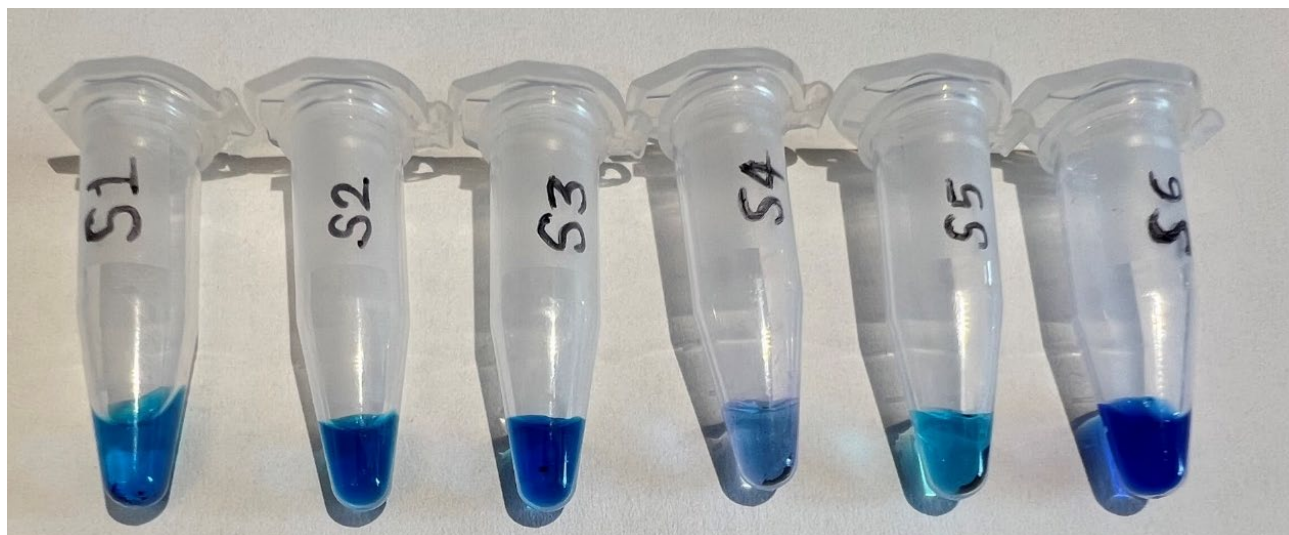


Figure S2. Methylene Blue test for perchlorates

Residual perchlorate ions are highly neurotoxic. To test for the presence of sodium perchlorate after ion-exchange HPLC, we adapted and used a methylene blue test. For this method, ion-exchange HPLC-purified Malat1 ASOs were aliquoted from the same stock and processed using different wash sequences (Kruger, D., and Tschirch, E. (1931). Detection of Small Quantities of Perchlorates. *Z. Anal. Chem* 85, 171-176). We used a 0.3% methylene blue solution, with 5 μ L of methylene blue solution added to each 100 μ L Malat1 ASO aliquot as well as control standards (0.0001 M, 0.001 or 0.01 M perchlorate ions in tubes S3 to S5, respectively). The mixtures were then centrifuged. In a sample that had undergone insufficient washing (**tube S1**), violet crystalline precipitates were visibly observed indicating trace perchlorates. This was also observed in the three controls spiked with increasing concentrations of sodium perchlorate (**tubes S3-S5**). No precipitates were observed in the ASO sample that was subjected to more rigorous washing (**tube S2**) or the PBS only methylene blue control (**tube S6**).

Sample and observation details

S1: Malat1 Without PBS wash:

We observed a crystalline violet precipitate indicating the presence of perchlorate ions

S2: Malat1 ASO after thorough PBS wash; No precipitate observed.

S3: 0.0001M perchlorate ions;

We observed a slight crystalline violet precipitate indicating the presence of low concentration perchlorate ions

S4: 0.001M perchlorate ions;

We observed a crystalline violet precipitate indicating the presence of perchlorate ions

S5: 0.01M perchlorate ions;

We observed a crystalline violet precipitate indicating the presence of perchlorate ions

S6: PBS only; No precipitate observed.

Table S1. Buffers containing both calcium and phosphate show the presence of micron-scale particles within hours after preparation, even after sterile filtration.

All buffers were filtered using a 0.2 µm syringe filter immediately after preparation. Buffers were tested for the presence of microscopic particles by dynamic light scattering either 3 hours (left) or 24 hours (right) after preparation and sterile filtration of buffer formulations containing divalent cations at the listed concentrations. Solutions were prepared and maintained at room temperature until analysis. All samples were evaluated using DTS0012 cuvettes in duplicate. n.a. = particles were not detected by DLS. PDI = polydispersity index.

Formulation	3 hours		24 hours	
	Z-Average (d. nm) mean ± SD	Pdl mean ± SD	Z-Average (d. nm) mean ± SD	Pdl mean ± SD
PBS (no divalent)	n.a.	n.a.	n.a.	n.a.
PBS +3.6mM Ca ²⁺	12615 ± 2355	0.710 ± 0.257	7580 ± 4	0.793 ± 0.154
PBS +3.6mM Mg ²⁺	n.a.	n.a.	n.a.	n.a.
PBS +8mM Ca ²⁺	11470 ± 594	0.793 ± 0.144	9210 ± 802	0.530 ± 0.204
PBS +8mM Mg ²⁺	n.a.	n.a.	n.a.	n.a.
aCSF (no divalent)	n.a.	n.a.	n.a.	n.a.
aCSF +3.6mM Ca ²⁺	n.a.	n.a.	n.a.	n.a.
aCSF +3.6mM Mg ²⁺	n.a.	n.a.	n.a.	n.a.
aCSF +8mM Ca ²⁺	n.a.	n.a.	n.a.	n.a.
aCSF +8mM Ca ²⁺	n.a.	n.a.	n.a.	n.a.

Table S2. Particle measurements from Elliotts B solution after heating or freezing.

A USP grade vial of Elliotts B that was stored in 25°C (room temperature) was obtained. 1mL aliquots from this original vial were then briefly heated to 100°C and then stored at room temperature or directly stored at room temperature or -80°C without heating. All solutions were returned to room temperature for analysis. Solutions were tested for the presence of microscopic particles by dynamic light scattering (DLS) (Zetasizer, Malvern Panalytical) using DTS0012 cuvettes in sample duplicates (i.e., sample 1 and sample 2). Each sample duplicate was also measured in technical duplicate (i.e., sample 1.1 and sample 1.2). DLS did not detect any particles from Elliotts B solution stored at room temperature. Perturbations in temperature such as heating to 100°C or even brief storage at -80°C can introduce particles of detectable size by DLS. n.a. = particles were not detected by DLS. PDI = polydispersity index.

Temperature conditions before testing (°C)	Formulation	Z-Average (d. nm)	Pdl	Mean Count Rate (kcps)
100	Elliotts B - sample 1.1	7726	0.7705	23.05
100	Elliotts B - sample 1.2	4879	1	20.51
100	Elliotts B - sample 2.1	14400	0.8203	14.77
100	Elliotts B - sample 2.2	4787	1	4.033
25	Elliotts B - sample 3.1	n.a.	n.a.	n.a.
25	Elliotts B - sample 3.2	n.a.	n.a.	n.a.
25	Elliotts B - sample 4.1	n.a.	n.a.	n.a.
25	Elliotts B - sample 4.2	n.a.	n.a.	n.a.
-80	Elliotts B - sample 5.1	n.a.	n.a.	n.a.
-80	Elliotts B - sample 5.2	n.a.	n.a.	n.a.
-80	Elliotts B - sample 6.1	n.a.	n.a.	n.a.
-80	Elliotts B - sample 6.2	19900	1	0.429

Table S3. Raw data from the EvADINT scoring assay for each mouse in Figure 1A

ASO						C9ORF72 Full PS					
Mouse ID						41	42	43	44	45	46
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	20	15	0	0	15
Hyperactivity or spasticity	15	10		5		5	0	0	10	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	12	8	0	0	12
Unstimulated Movement	0	3	6	9	15	6	15	9	9	3	9
Movement without Ataxia	0	2	4	6	10	6	10	10	4	4	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	5	3	3	2	3
Total Score						20	62	45	26	9	49

ASO						C9ORF72 PO/PS					
Mouse ID						53	54	55	56	57	58
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		10	0	0	5	5	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	8	0
Unstimulated Movement	0	3	6	9	15	9	0	6	6	9	3
Movement without Ataxia	0	2	4	6	10	4	4	6	6	6	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	2	2	2	5	1
Total Score						28	6	14	19	33	8

ASO						PBS							
Mouse ID						33	34	35	36	37	38	39	40
Death	75					0	0	0	0	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h								
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	0	0	0	0	0	0	0	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	1	0	1	1	1	1	0
Total Score						1	1	0	1	1	1	1	2

Table S4. Raw data from the EvADINT scoring assay for each mouse in Figure 1B

ASO						Malat1 Full PS					
Mouse ID						105	106	107	108	109	110
Death	75					0	0	0	75	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		15	0	0	0	10	10
Hyperactivity or spasticity	15	10		5		0	10	10	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	20	0	20	0	20	20
Unstimulated Movement	0	3	6	9	15	15	3	15	0	15	15
Movement without Ataxia	0	2	4	6	10	10	6	10	0	10	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	1	5	0	5	5
Total Score						65	20	60	75	60	60

ASO						Malat1 PO/PS					
Mouse ID						117	118	119	120	121	122
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		5	5	5	5	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	12	12	12	12	8	8
Unstimulated Movement	0	3	6	9	15	9	9	9	9	6	9
Movement without Ataxia	0	2	4	6	10	10	10	10	10	10	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	5	3	5	2	3
Total Score						39	41	39	41	26	30

ASO						PBS					
Mouse ID						129	130	131	132	133	134
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	2	2	2	2	2	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	1	1	1	1	1
Total Score						3	3	3	3	3	3

Table S5. Raw data from the EvADINT scoring assay for each mouse in Figure 1C

ASO						Htt Full PS							
Mouse ID						1	2	3	4	5	6	7	8
Death	75					0	0	0	0	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	15	0	0	15
Hyperactivity or spasticity	15	10		5		0	10	10	10	0	0	10	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h								
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	20	12	4	20
Unstimulated Movement	0	3	6	9	15	9	9	9	9	15	9	6	15
Movement without Ataxia	0	2	4	6	10	6	10	10	10	10	3	6	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	3	3	3	5	3	2	5
Total Score						20	32	32	32	65	27	28	65

ASO						Htt PO/PS							
Mouse ID						17	18	19	20	21	22	23	24
Death	75					0	75	0	0	0	0	0	
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	20	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0	15	5
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h								
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	12	4	0
Unstimulated Movement	0	3	6	9	15	6	0	0	9	0	15	9	3
Movement without Ataxia	0	2	4	6	10	6	0	4	10	4	10	6	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	0	3	1	0	5	3	2
Total Score						17	75	7	20	4	62	37	14

ASO						PBS							
Mouse ID						33	34	35	36	37	38	39	40
Death	75					0	0	0	0	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h								
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	0	0	0	0	0	0	0	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	1	0	1	1	1	1	0
Total Score						1	1	0	1	1	1	1	2

Table S6. Raw data from the EvADINT scoring assay for each mouse in Figure 1D

ASO						NTC Full PS					
Mouse ID						135	136	137	138	139	140
Death	75					75	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	10	10	10	10	5
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	20	20	0	20	0
Unstimulated Movement	0	3	6	9	15	0	15	15	9	15	3
Movement without Ataxia	0	2	4	6	10	0	10	10	6	10	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	5	5	3	5	1
Total Score						75	60	60	28	60	15

ASO						NTC PO/PS					
Mouse ID						147	148	149	150	151	152
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	4	4	4	4	4	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	2	2	2	0	1
Total Score						6	6	6	6	4	5

ASO						PBS					
Mouse ID						129	130	131	132	133	134
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	2	2	2	2	2	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	1	1	1	1	1
Total Score						3	3	3	3	3	3

Table S7. Raw data from the EvADINT scoring assay for each mouse in Figure 2

ASO						PS DNA							
Mouse ID						65	66	67	68	69	70	71	72
Death	75					0	0	0	0	0	0	0	75
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		20	0	20	20	20	15	10	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0	0	0
	≥24												
Time required for:	0.5 h	1 h	2 h	4 h	h								
Maintenance of sternal posture	0	4	8	12	20	20	0	20	20	20	4	4	0
Unstimulated Movement	0	3	6	9	15	15	9	15	15	15	6	15	0
Movement without Ataxia	0	2	4	6	10	10	10	10	10	10	6	10	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	5	5	5	5	5	5	0
Total Score						70	24	70	70	70	36	44	75

ASO						PS 2'-O-MOE							
Mouse ID						73	74	75	76	77	78	79	80
Death	75					0	0	0	0	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	10	10	0
Hyperactivity or spasticity	15	10		5		10	10	10	10	10	0	0	10
	≥24												
Time required for:	0.5 h	1 h	2 h	4 h	h								
Maintenance of sternal posture	0	4	8	12	20	4	4	4	4	4	0	0	4
Unstimulated Movement	0	3	6	9	15	9	9	9	6	9	0	0	9
Movement without Ataxia	0	2	4	6	10	6	6	6	6	6	6	6	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	3	3	3	3	3	3	3
Total Score						32	32	32	29	32	19	19	32

ASO						PS 2'-O-Me							
Mouse ID						81	82	83	84	85	86	87	88
Death	75					0	0	0	0	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	0	0	10
Hyperactivity or spasticity	15	10		5		5	10	10	15	10	5	5	0
	≥24												
Time required for:	0.5 h	1 h	2 h	4 h	h								
Maintenance of sternal posture	0	4	8	12	20	8	4	0	8	4	0	0	0
Unstimulated Movement	0	3	6	9	15	6	3	3	9	3	0	3	6
Movement without Ataxia	0	2	4	6	10	6	6	4	6	6	6	6	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	3	3	3	3	5	3	3
Total Score						28	26	20	41	26	16	17	29

ASO						PBS							
Mouse ID						89	90	91	92	93	94	95	96
Death	75					0	0	0	0	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	5	0	0	0	0	0
	≥24												
Time required for:	0.5 h	1 h	2 h	4 h	h								
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	2	2	4	4	4	2	2	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	1	0	0	0	0	0
Total Score						2	2	10	4	4	2	2	4

Table S8. Raw data from the EvADINT scoring assay for each mouse in Figure 3a

ASO						B6 WT / PBS		
Mouse ID						6769	6770	6751
Death	75					0	0	0
	Severe			Moderate	Mild			
Tonic seizure	20			15	10	0	0	0
Hyperactivity or spasticity	15			10	5	0	0	0
						≥24		
Time required for:	0.5 h	1 h	2 h	4 h	h			
Maintenance of sternal posture	0	4	8	12	20	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0
Movement without Ataxia	0	2	4	6	10	0	0	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	1	1
Total Score						1	1	1

ASO						UNC93B1 / PBS		
Mouse ID						6752	6753	6754
Death	75					0	0	0
	Severe			Moderate	Mild			
Tonic seizure	20			15	10	0	0	0
Hyperactivity or spasticity	15			10	5	0	0	0
						≥24		
Time required for:	0.5 h	1 h	2 h	4 h	h			
Maintenance of sternal posture	0	4	8	12	20	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0
Movement without Ataxia	0	2	4	6	10	2	2	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	2	2
Total Score						4	4	4

ASO						B6 WT / Full DNA		
Mouse ID						6758	6759	6760
Death	75					75	75	75
	Severe			Moderate	Mild			
Tonic seizure	20			15	10	0	0	0
Hyperactivity or spasticity	15			10	5	0	0	0
						≥24		
Time required for:	0.5 h	1 h	2 h	4 h	h			
Maintenance of sternal posture	0	4	8	12	20	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0
Movement without Ataxia	0	2	4	6	10	0	0	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	0
Total Score						75	75	75

ASO						UNC93B1 / Full DNA		
Mouse ID						6755	6756	6757
Death	75					75	75	75
	Severe		Moderate		Mild			
Tonic seizure	20		15		10	0	0	0
Hyperactivity or spasticity	15		10		5	0	0	0
					≥24			
Time required for:	0.5 h	1 h	2 h	4 h				
Maintenance of sternal posture	0	4	8	12	20	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0
Movement without Ataxia	0	2	4	6	10	0	0	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	0
Total Score						75	75	75

Table S9. Raw data from the EvADINT scoring assay for each mouse in Figure 3b

ASO						B6 WT / PBS				
Mouse ID						159	160	161	162	163
Death	75					0	0	0	0	0
	Severe	Moderate		Mild						
Tonic seizure	20	15		10		0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h					
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	4	4	4	4	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	2	2	2	2
Total Score						6	6	6	6	6

ASO						B6 WT / Full DNA				
Mouse ID						164	165	166	167	168
Death	75					75	0	75	0	75
	Severe	Moderate		Mild						
Tonic seizure	20	15		10		0	0	0	10	0
Hyperactivity or spasticity	15	10		5		0	15	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h					
Maintenance of sternal posture	0	4	8	12	20	0	20	0	20	0
Unstimulated Movement	0	3	6	9	15	0	15	0	15	0
Movement without Ataxia	0	2	4	6	10	0	10	0	10	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	5	0	5	0
Total Score						75	65	75	60	75

ASO						MyD88.STING / PBS		
Mouse ID						169	170	171
Death	75					0	0	0
	Severe	Moderate		Mild				
Tonic seizure	20	15		10		0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h			
Maintenance of sternal posture	0	4	8	12	20	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0
Movement without Ataxia	0	2	4	6	10	4	4	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	2	1
Total Score						5	6	5

ASO						MyD88.Sting / Full DNA			
Mouse ID						172	173	174	175
Death	75					75	75	75	0
	Severe		Moderate		Mild				
Seizure	20		15		10	0	0	0	10
Hyperactivity or spasms	15		10		5	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h				
Maintenance of sternal posture	0	4	8	12	20	0	0	0	20
Unstimulated Movement	0	3	6	9	15	0	0	0	15
Movement without Ataxia	0	2	4	6	10	0	0	0	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	0	5
Total Score						75	75	75	60

Table S10. Raw data from the EvADINT scoring assay for each mouse in Figure 4a

ASO						C9ORF72 Full PS w/Ca2+					
Mouse ID						47	48	49	50	51	52
Death						0	0	0	0	0	0
	Severe	75		Mild							
		Moderate									
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		10	0	5	5	0	0
				≥24							
Time required for:	0.5 h	1 h	2 h	4 h	h						
Maintenance of sternal posture	0	4	8	12	20	8	0	12	8	0	12
Unstimulated Movement	0	3	6	9	15	9	0	9	9	0	9
Movement without Ataxia	0	2	4	6	10	6	4	10	10	2	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	3	5	5	0	5
Total Score						36	7	41	37	2	36

ASO						C9ORF72 PO/PS w/Ca2+					
Mouse ID						59	60	61	62	63	64
Death						0	0	0	0	0	0
	Severe	75		Mild							
		Moderate									
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0
				≥24							
Time required for:	0.5 h	1 h	2 h	4 h	h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	3	3	6	0	0
Movement without Ataxia	0	2	4	6	10	0	6	6	6	6	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	1	2	2	1	1
Total Score						2	10	11	14	7	7

Table S11. Raw data from the EvADINT scoring assay for each mouse in Figure 4b

ASO					Malat1 Full PS w/Ca2+					
Mouse ID					111	112	113	114	115	116
Death	75				0	0	0	0	0	0
	Severe	Moderate		Mild						
Tonic seizure	20	15		10	10	10	10	10	10	10
Hyperactivity or spasticity	15	10		5	0	0	0	0	0	0
	0.5 h	1 h	2 h	4 h	≥24 h					
Time required for:										
Maintenance of sternal posture	0	4	8	12	20	12	20	0	20	20
Unstimulated Movement	0	3	6	9	15	9	15	9	15	15
Movement without Ataxia Normal	0	2	4	6	10	10	10	10	10	10
Grooming/Eating/Nesting	0	1	2	3	5	3	5	3	5	5
Total Score						44	60	32	60	60

ASO					Malat1 PO/PS w/Ca2+					
Mouse ID					123	124	125	126	127	128
Death	75				0	0	0	0	0	0
	Severe	Moderate		Mild						
Tonic seizure	20	15		10	0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5	5	0	0	0	0	0
	0.5 h	1 h	2 h	4 h	≥24 h					
Time required for:										
Maintenance of sternal posture	0	4	8	12	20	0	0	8	0	12
Unstimulated Movement	0	3	6	9	15	9	6	9	6	9
Movement without Ataxia Normal	0	2	4	6	10	6	10	10	10	4
Grooming/Eating/Nesting	0	1	2	3	5	3	2	3	3	1
Total Score						23	18	30	19	5

Table S12. Raw data from the EvADINT scoring assay for each mouse in Figure 4c

ASO						Htt Full PS w/Ca2+							
Mouse ID						9	10	11	12	13	14	15	16
Death	75					0	0	0	0	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0	0	0
	0.5 h	1 h	2 h	4 h	≥24 h								
Time required for:													
Maintenance of sternal posture	0	4	8	12	20	0	4	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	9	0	3	6	3	6	3
Movement without Ataxia	0	2	4	6	10	0	10	10	10	10	6	2	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	5	0	1	2	2	6	2
Total Score						3	28	10	14	18	11	14	9

ASO						Htt PO/PS w/Ca2+							
Mouse ID						25	26	27	28	29	30	31	32
Death	75					0	0	0	75	0	0	0	0
	Severe	Moderate		Mild									
Tonic seizure	20	15		10		0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0	0	0
	0.5 h	1 h	2 h	4 h	≥24 h								
Time required for:													
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	3	3	6	0	3	0	0	0
Movement without Ataxia	0	2	4	6	10	6	10	10	0	10	2	2	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	2	2	0	1	0	0	0
Total Score						12	15	18	75	14	2	2	2

Table S13. Raw data from the EvADINT scoring assay for each mouse in Figure 4d

ASO						NTC Full PS w/Ca2+					
Mouse ID						141	142	143	144	145	146
Death	75					0	0	0	0	0	75
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		10	10	10	10	10	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	12	12	12	12	12	0
Unstimulated Movement	0	3	6	9	15	15	15	9	9	9	0
Movement without Ataxia	0	2	4	6	10	10	10	10	10	10	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	5	5	5	3	0
Total Score						0	52	46	46	44	75

ASO						NTC PO/PS w/Ca2+					
Mouse ID						153	154	155	156	157	158
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	4	4	4	4	4	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	1	2	1	1
Total Score						4	4	5	6	5	5

Table S14. Raw data from the EvADINT scoring assay for each mouse in Figure 5. Part 1/7

ASO						PBS									
Mouse ID						8249	8250	8253	8256	8259	8931	8932	8133	8134	8135
Death	75					0	0	0	0	0	0	0	0	0	0
	Severe	Moderate	Mild												
Tonic seizure	20	15	10				0	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10	5				0	0	0	0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h										
Maintenance of sternal posture	0	4	8	12	20	8	0	0	12	8	0	12	0	12	0
Unstimulated Movement	0	3	6	9	15	6	0	0	9	6	0	9	0	9	0
Movement without Ataxia	0	2	4	6	10	6	6	6	6	6	4	10	2	6	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	3	1	5	3	3	5	5	3	3
Total Score						23	9	7	32	23	7	36	7	30	7

ASO						PBS (8Ca)				
Mouse ID						8246	8247	8252	8255	8258
Death	75					0	0	0	0	0
	Severe	Moderate	Mild							
Tonic seizure	20	15	10				0	0	0	0
Hyperactivity or spasticity	15	10	5				0	0	0	0
	Time required for:									
	0.5 h	1 h	2 h	4 h	≥24 h					
Maintenance of sternal posture	0	4	8	12	20	0	12	12	12	20
Unstimulated Movement	0	3	6	9	15	6	9	6	6	15
Movement without Ataxia	0	2	4	6	10	4	6	6	6	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	3	5	5	5
Total Score						13	30	29	29	50

Table S14 continued (part 2/7).

ASO						PBS (8Mg)									
Mouse ID						8236	8239	8240	8242	8243	8902	8905	8908	8911	8914
Death	75					0	0	0	0	0	0	0	0	0	0
	Severe	Moderate		Mild											
Tonic seizure	20	15	10			0	10	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10	5			0	0	0	0	0	0	0	0	0	0
Time required for: 0.5 h 1 h 2 h 4 h ≥24 h															
Maintenance of sternal posture	0	4	8	12	20	8	0	0	4	20	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	6	0	0	6	15	0		0	0	3
Movement without Ataxia	0	2	4	6	10	6	6	6	6	10	2	0	2	0	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	5	3	3	5	2	5	3	3	2
Total Score						25	21	9	19	50	4	5	5	3	7

ASO						PBS (3.6Mg)				
Mouse ID						8939	8941	8944	8947	8950
Death	75					0	0	0	0	0
	Severe	Moderate		Mild						
Tonic seizure	20	15		10		0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0
Time required for: 0.5 h 1 h 2 h 4 h ≥24 h										
Maintenance of sternal posture	0	4	8	12	20	0	8	8	4	0
Unstimulated Movement	0	3	6	9	15	0	6	6	3	0
Movement without Ataxia	0	2	4	6	10	2	6	6	4	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	5	5	3	3
Total Score						4	25	25	14	9

Table S14 continued (part 3/7).

ASO						PBS (empty)									
Mouse ID						8261	8262	8263	8264	8265	8936	8939	8942	8945	8948
Death	75					0	0	0	0	0	0	0	0	0	0
	Severe	Moderate	Mild												
Tonic seizure	20	15	10			0	0	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10	5			0	0	0	0	0	0	0	0	0	0
Time required for: 0.5 h 1 h 2 h 4 h ≥24 h															
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	0	0	0	0	0	2	0	0	0	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	0	1	0	0	1	1	1	0	1
Total Score						1	0	1	0	0	3	1	1	0	1

ASO						aCSF									
Mouse ID						8201	8204	8207	8210	8213	8903	8906	8909	8912	8915
Death	75					0	0	0	0	0	0	0	0	0	0
	Severe	Moderate	Mild												
Tonic seizure	20	15	10			0	0	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10	5			0	0	0	0	0	0	0	0	0	0
Time required for: 0.5 h 1 h 2 h 4 h ≥24 h															
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	4	0	4
Unstimulated Movement	0	3	6	9	15	0	0	0	2	0	0	0	3	3	6
Movement without Ataxia	0	2	4	6	10	2	0	0	0	0	0	0	4	2	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	5	3	5	3	1	2	3	5	3
Total Score						7	5	3	7	3	1	2	14	10	19

Table S14 continued (part 4/7).

ASO					aCSF (8Ca)									
Mouse ID					8221	8224	8227	8230	8233	8937	8940	8943	8946	8949
Death	75				0	0	0	0	0	0	0	0	0	0
	Severe	Moderate	Mild											
Tonic seizure	20	15	10		10	0	0	0	10	0	0	0	0	0
Hyperactivity or spasticity	15	10	5		0	0	0	0	0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h									
Maintenance of sternal posture	0	4	8	12	20	4	0	0	0	0	8	8	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	6	6	0	6
Movement without Ataxia	0	2	4	6	10	4	0	0	0	2	6	4	6	6
Normal Grooming/Eating/Nesting	0	1	2	3	5					2	3	3	5	3
Total Score						3	3	2	3	2	4	23	21	15

ASO					aCSF (8Mg)									
Mouse ID					8222	8225	8228	8231	8234	8901	8904	8907	8910	8913
Death	75				0	0	0	0	0	0	0	0	0	0
	Severe	Moderate	Mild											
Tonic seizure	20	15	10		0	0	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10	5		0	0	0	0	0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h									
Maintenance of sternal posture	0	4	8	12	20	4	0	0	0	4	12	4	0	12
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	9	9	0	9
Movement without Ataxia	0	2	4	6	10	4	4	4	4	4	6	10	0	6
Normal Grooming/Eating/Nesting	0	1	2	3	5					5	5	3	3	5
Total Score						3	3	3	3	3	11	32	28	3

Table S14 continued (part 5/7).

ASO						aCSF (5Ca, 3Mg)				
Mouse ID						8202	8205	8208	8211	8214
Death	75					0	0	0	0	0
	Severe	Moderate		Mild						
Tonic seizure	20	15		10		0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0
Time required for: 0.5 h 1 h 2 h 4 h ≥24 h										
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	4	2	2	2	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	1	3	3	5
Total Score						9	3	5	5	5

ASO						aCSF (3.6Ca)									
Mouse ID						8203	8206	8209	8212	8215	8916	8919	8922	8925	8928
Death	75					0	0	0	0	0	0	0	0	0	0
	Severe	Moderate		Mild											
Tonic seizure	20	15		10		0	0	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0	0	0	0	0
Time required for: 0.5 h 1 h 2 h 4 h ≥24 h															
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	6	6	6	4	4	2	2	0	2	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	5	5	5	3	5	2	2	2	2	3
Total Score						11	11	11	7	9	4	4	2	4	5

Table S14 continued (part 6/7).

ASO						aCSF (3.6Mg)									
Mouse ID						8216	8217	8218	8219	8220	8957	8952	8953	8954	8955
Death	75					0	0	0	0	0	0	0	0	0	0
	Severe	Moderate	Mild												
Tonic seizure	20	15	10			0	0	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10	5			0	0	0	0	0	0	0	0	0	0
Time required for:						0.5 h	1 h	2 h	4 h	≥24 h					
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	12	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0	0	9	0	0
Movement without Ataxia	0	2	4	6	10	2	0	0	0	0	6	6	6	0	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	3	3	2	2	5	5	5	5	5
Total Score						2	3	3	2	2	11	11	32	5	7

ASO						aCSF (empty)									
Mouse ID						8248	8251	8254	8257	8260	8918	8921	8924	8927	8930
Death	75					0	0	0	0	0	0	0	0	0	0
	Severe	Moderate	Mild												
Tonic seizure	20	15	10				0	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15	10	5				0	0	0	0	0	0	0	0	0
Time required for: 0.5 h 1 h 2 h 4 h ≥24 h															
Maintenance of sternal posture	0	4	8	12	20	0	0	0	8	8	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	6	6	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	2	4	0	4	4	0	0	0	0	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	3	3	3	3	2	0	1	2	1
Total Score						5	7	3	21	21	2	0	1	2	3

Table S14 continued (part 7/7).

ASO						LRS				
Mouse ID						8917	8920	8923	8926	8929
Death	75					0	0	0	0	0
	Severe	Moderate		Mild						
Tonic seizure	20	15		10		0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h					
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	0	2	0	0	4
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	3	2	1	2
Total Score						2	5	2	1	6

ASO						LRS (empty)				
Mouse ID						35	38	41	44	47
Death	75					0	0	0	0	0
	Severe	Moderate		Mild						
Tonic seizure	20	15		10		0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h					
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	3	0	0	0
Movement without Ataxia	0	2	4	6	10	0	2	0	0	0
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	0	2	2	2
Total Score						2	5	2	2	2

Table S15. Raw data from the EvADINT scoring assay for each mouse in Figure S1a. (part 1/2)

ASO						C9orf72 full PS w/Ca2+ (35nmol)					
Mouse ID						47	48	49	50	51	52
Death						75	0	0	0	0	0
	Severe		Moderate		Mild						
Tonic seizure	20		15		10	0	0	0	0	0	0
Hyperactivity or spasticity	15		10		5	10	0	5	5	0	0
					≥24 h						
Time required for:	0.5 h	1 h	2 h	4 h							
Maintenance of sternal posture	0	4	8	12	20	8	0	12	8	0	12
Unstimulated Movement	0	3	6	9	15	9	0	9	9	0	9
Movement without Ataxia	0	2	4	6	10	6	4	10	10	2	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	3	5	5	0	5
Total Score						36	7	41	37	2	36

ASO						C9orf72 full PS w/Ca2+ (15nmol)					
Mouse ID						176	177	178	179	180	181
Death						75	0	0	0	0	0
	Severe		Moderate		Mild						
Tonic seizure	20		15		10	0	0	0	0	0	0
Hyperactivity or spasticity	15		10		5	0	0	0	0	0	0
					≥24 h						
Time required for:	0.5 h	1 h	2 h	4 h							
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	6	6	6	6	6	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	1	1	1	0	1
Total Score						6	7	7	7	6	7

ASO						C9orf72 PO/PS w/Ca2+ (35nmol)					
Mouse ID						25	26	27	28	29	30
Death						75	0	0	0	0	0
	Severe		Moderate		Mild						
Tonic seizure	20		15		10	0	0	0	0	0	0
Hyperactivity or spasticity	15		10		5	0	0	0	0	0	0
					≥24 h						
Time required for:	0.5 h	1 h	2 h	4 h							
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	3	3	6	0	0
Movement without Ataxia	0	2	4	6	10	0	6	6	6	6	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	1	2	2	1	1
Total Score						2	10	11	14	7	7

Table S15 continued (part 2/2).

ASO						C9orf72 PO/PS w/Ca2+ (15nmol)					
Mouse ID						182	183	184	185	186	187
Death						75	0	0	0	0	0
	Severe	Moderate		Mild							
Tonic seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasticity	15	10		5		0	0	0	0	0	0
				≥24 h							
Time required for:	0.5 h	1 h	2 h	4 h							
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	6	6	6	6	6	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	0	0	0	0
Total Score						6	6	6	6	6	6

ASO						PBS							
Mouse ID						33	34	35	36	37	38	39	40
Death						75	0	0	0	0	0	0	0
	Severe		Moderate		Mild								
Tonic seizure	20		15		10	0	0	0	0	0	0	0	0
Hyperactivity or spasticity	15		10		5	0	0	0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h								
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	0	0	0	0	0	0	0	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	1	1	0	1	1	1	1	0
Total Score						1	1	0	1	1	1	1	2

Table S16. Raw data from the EvADINT scoring assay for each mouse in Figure S1b (part 1/2)

ASO						<i>Malat1</i> Full PS w/Ca2+ (35nmol)					
Mouse ID						111	112	113	114	115	116
Death						75	0	0	0	0	0
	Severe		Moderate		Mild						
Tonic seizure	20		15		10	10	10	10	10	10	10
Hyperactivity or spasticity	15		10		5	0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	12	20	0	20	20	20
Unstimulated Movement	0	3	6	9	15	9	15	9	15	15	15
Movement without Ataxia	0	2	4	6	10	10	10	10	10	10	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	5	3	5	5	5
Total Score						44	60	32	60	60	60

ASO						<i>Malat1</i> Full PS w/Ca2+ (15nmol)					
Mouse ID						188	189	190	191	192	193
Death						75	0	0	0	0	0
	Severe		Moderate		Mild						
Seizure	20		15		10	0	10	10	0	10	10
Hyperactivity or spasms	15		10		5	10	0	0	10	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	8	12	12	12	12
Unstimulated Movement	0	3	6	9	15	0	9	15	15	15	15
Movement without Ataxia	0	2	4	6	10	10	10	10	10	10	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	2	5	5	5	5	5
Total Score						22	42	52	52	52	52

ASO						<i>Malat1</i> PO/PS w/Ca2+ (35nmol)					
Mouse ID						123	124	125	126	127	128
Death						75	0	0	0	0	0
	Severe		Moderate		Mild						
Seizure	20		15		10	0	0	0	0	0	0
Hyperactivity or spasms	15		10		5	5	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	8	0	0	12
Unstimulated Movement	0	3	6	9	15	9	6	9	6	0	9
Movement without Ataxia	0	2	4	6	10	6	10	10	10	4	10
Normal Grooming/Eating/Nesting	0	1	2	3	5	3	2	3	3	1	3
Total Score						23	18	30	19	5	34

Table S16 continued (part 2/2).

ASO						<i>Malat1</i> PO/PS w/Ca2+ (15nmol)					
Mouse ID						194	195	196	197	198	199
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasms	15	10		5		0	0	0	5	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	3	0	0
Movement without Ataxia	0	2	4	6	10	6	6	6	6	6	6
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	0	2	1	1
Total Score						6	6	6	16	7	7

ASO						PBS					
Mouse ID						200	201	202	203	204	205
Death	75					0	0	0	0	0	0
	Severe	Moderate		Mild							
Seizure	20	15		10		0	0	0	0	0	0
Hyperactivity or spasms	15	10		5		0	0	0	0	0	0
Time required for:	0.5 h	1 h	2 h	4 h	≥24 h						
Maintenance of sternal posture	0	4	8	12	20	0	0	0	0	0	0
Unstimulated Movement	0	3	6	9	15	0	0	0	0	0	0
Movement without Ataxia	0	2	4	6	10	2	2	2	4	2	2
Normal Grooming/Eating/Nesting	0	1	2	3	5	0	0	0	1	0	0
Total Score						2	2	2	5	2	2